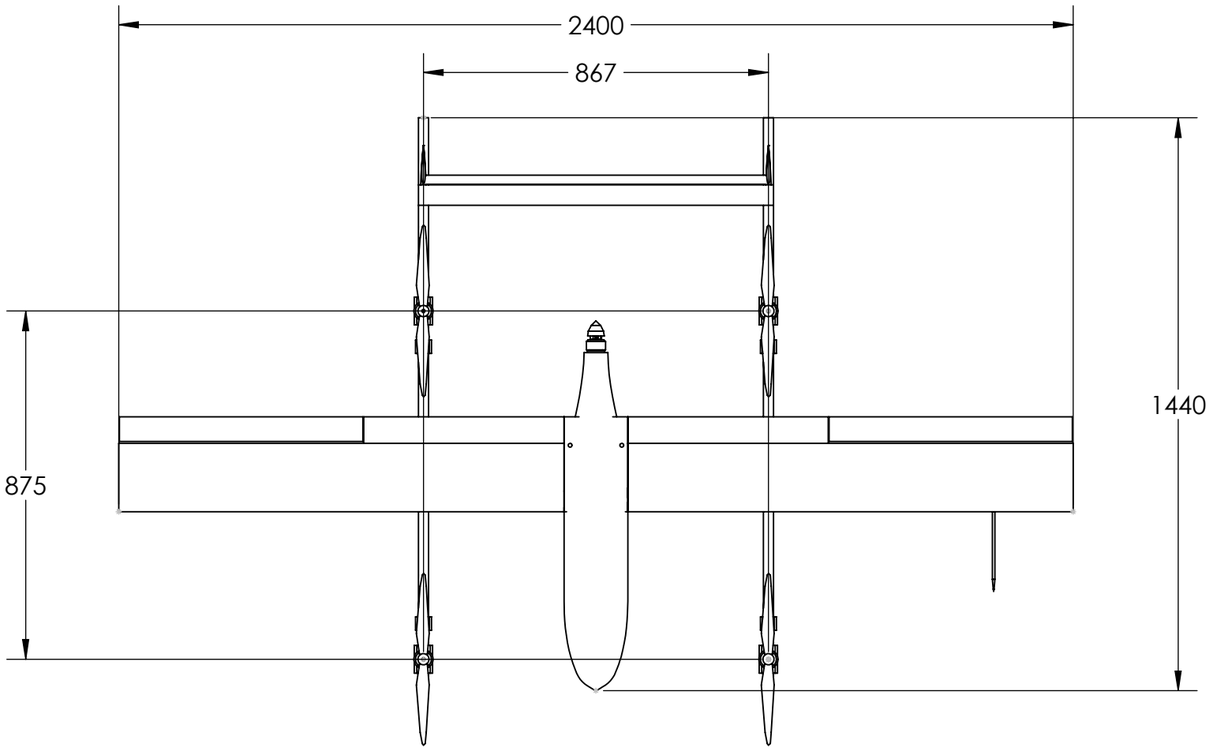
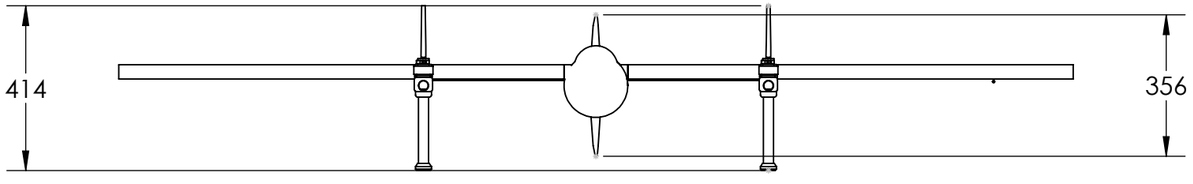
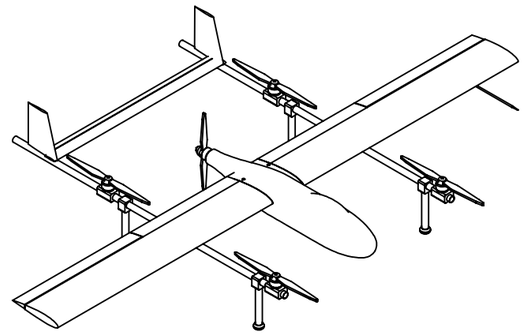


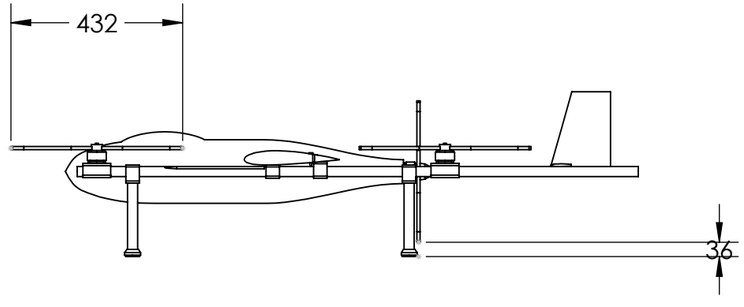
PLAN



FRONT ELEVATION



ISOMETRIC



SIDE ELEVATION

NOTES
1. DIMENSIONS IN MM. VIEWS 1:15, THIRD-ANGLE; ISOMETRIC 1:22.
2. PROPS SHOWN PARKED FORE-AFT (NOTIONAL): THE ESCs CANNOT PARK THEM.
3. SIZES ARE THE SIZED CONCEPT (MTOW 6.18 kg), NOT A BUILT ARTICLE.

DESIGN-BUILD-FLY

STUDENT CAPSTONE PROJECT, UC BERKELEY.
CONCEPT GEOMETRY FOR DESIGN REVIEW,
GENERATED FROM analysis/results.json
AND cad/params.txt (v0.3.1).
NOT RELEASED FOR MANUFACTURE.

UNLESS OTHERWISE SPECIFIED:		
DIMENSIONS ARE IN MILLIMETRES GENERAL TOLERANCES (EST): LINEAR X ±1 X.X ±0.5 ANGULAR ±1° PRINTED PARTS ±0.3		
INTERPRET GEOMETRIC TOLERANCING PER:		
MATERIAL SEE PARTS LIST		
FINISH SEE PARTS LIST	NEXT ASSY	USED ON
DO NOT SCALE DRAWING	APPLICATION	

	NAME	DATE
DRAWN	DBF	2026-10-18
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

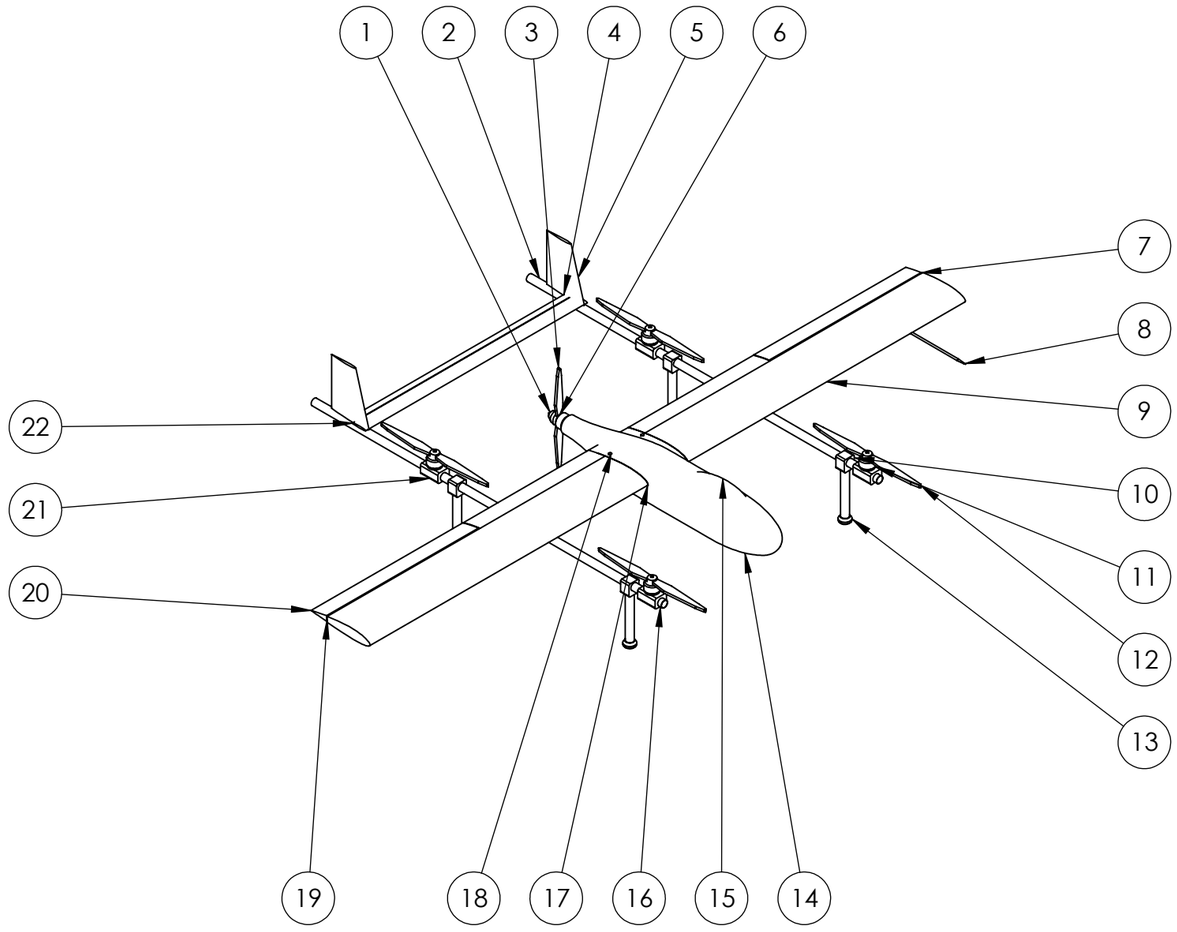
DESIGN-BUILD-FLY

TITLE:
LIFT + CRUISE VTOL
GENERAL ARRANGEMENT

SIZE	DWG. NO.	REV
B	DBF-DRW-001	0.3.1

SCALE: 1:15 WEIGHT: SHEET 1 OF 5

SOLIDWORKS Educational Product. For Instructional Use Only.



ITEM NO.	PART NUMBER	QTY.
1	Spinner	1
2	Boom_Rear_Stub	2
3	Pusher_Prop_14in	1
4	Elevator	1
5	Fin	2
6	Pusher_Motor	1
7	Aileron_L	1
8	Pitot_Tube	1
9	Wing_Panel_L	1
10	Gear_Clamp	4
11	Lift_Motor	4
12	Lift_Prop	4
13	Landing_Leg	4
14	Fuselage_Pod	1
15	Canopy	1
16	Boom_Front	2
17	Wing_Saddle	1
18	Wing_Bolt	2
19	Wing_Panel_R	1
20	Aileron_R	1
21	Motor_Mount	4
22	H_Tail	1
23	Spar_Tube	2
24	Wing_Joiner	1
25	Battery_6S_LiPo	1
26	Boom_Clamp_Front	2
27	Boom_Clamp_Rear	2

NOTES
1. ITEM NUMBERS MATCH THE CAD ASSEMBLY.
2. PURCHASED ITEMS, PRICES AND VENDORS: docs/purchasing/C0_BOM.csv.

DESIGN-BUILD-FLY

STUDENT CAPSTONE PROJECT, UC BERKELEY.
CONCEPT GEOMETRY FOR DESIGN REVIEW,
GENERATED FROM analysis/results.json
AND cad/params.txt (v0.3.1).
NOT RELEASED FOR MANUFACTURE.

		UNLESS OTHERWISE SPECIFIED:
		DIMENSIONS ARE IN MILLIMETRES GENERAL TOLERANCES (EST): LINEAR X ±1 X.X ±0.5 ANGULAR ±1° PRINTED PARTS ±0.3
		INTERPRET GEOMETRIC TOLERANCING PER:
		MATERIAL SEE PARTS LIST
NEXT ASSY	USED ON	FINISH SEE PARTS LIST
APPLICATION		DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	DBF	2026-10-18
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

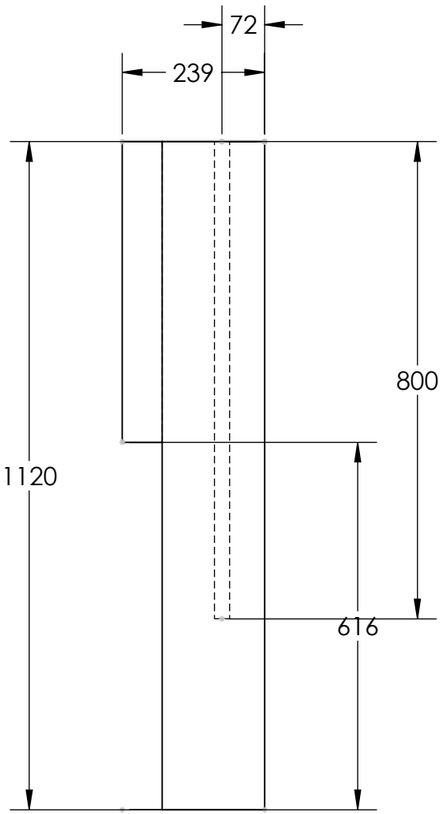
DESIGN-BUILD-FLY

TITLE:
LIFT + CRUISE VTOL
PARTS LIST

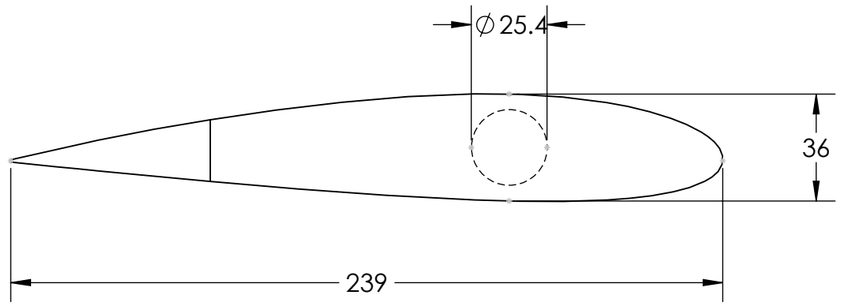
SIZE	DWG. NO.	REV
B	DBF-DRW-001	0.3.1

SCALE: 1:17 | WEIGHT: | SHEET 2 OF 5

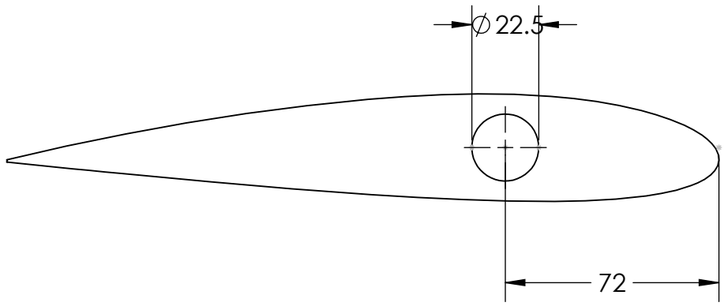
SOLIDWORKS Educational Product. For Instructional Use Only.



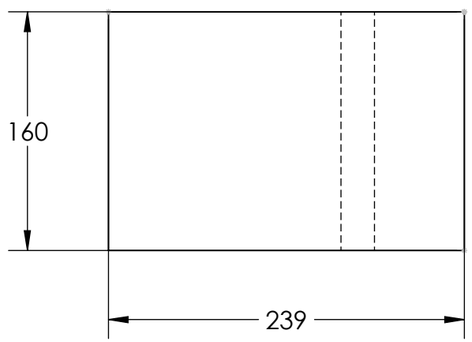
WING PANEL, RIGHT (LEFT IS MIRROR) 1:10



ROOT SECTION 1:2



SADDLE SECTION 1:2



WING SADDLE 1:4

NOTES
1. NACA 2415 (t/c FROM SIZING). XPS CORE, PACKING-TAPE SKIN.
2. AILERON CUT-OUT: AFT 28 % OF CHORD, OUTER 55 % OF SPAN. AILERON 26 % CHORD WITH A 2 % HINGE GAP.
3. SPAR SOCKET: 1.000 x 0.058 in 6061-T6, BONDED 800 mm INTO THE ROOT AT 30 % CHORD.
4. SADDLE: XPS BLOCK, PRINTED PETG ROOT PLATES; BORE FOR THE 0.875 in JOINER (0.23 mm CLEARANCE).

DESIGN-BUILD-FLY
STUDENT CAPSTONE PROJECT, UC BERKELEY.
CONCEPT GEOMETRY FOR DESIGN REVIEW,
GENERATED FROM analysis/results.json
AND cad/params.txt (v0.3.1).
NOT RELEASED FOR MANUFACTURE.

NEXT ASSY	USED ON	
APPLICATION		

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETRES
GENERAL TOLERANCES (EST):
LINEAR X ±1 X.X ±0.5
ANGULAR ±1°
PRINTED PARTS ±0.3
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL: SEE PARTS LIST
FINISH: SEE PARTS LIST
DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	DBF	2026-10-18
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

DESIGN-BUILD-FLY

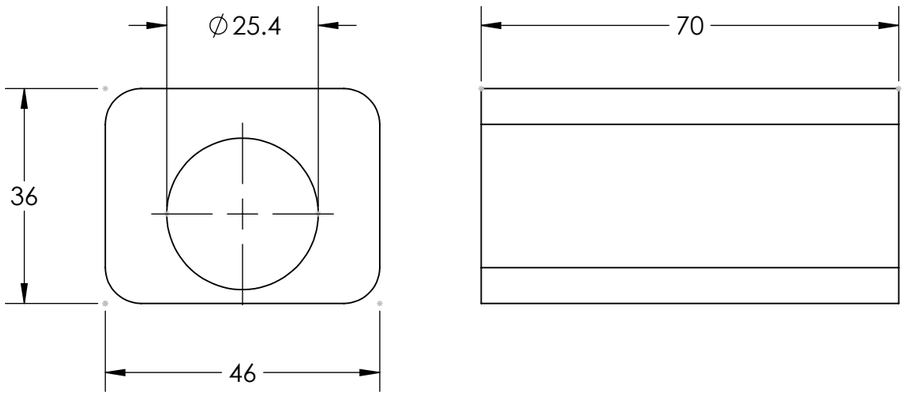
TITLE:
WING PANEL AND WING SADDLE

SIZE	DWG. NO.	REV
B	DBF-DRW-001	0.3.1

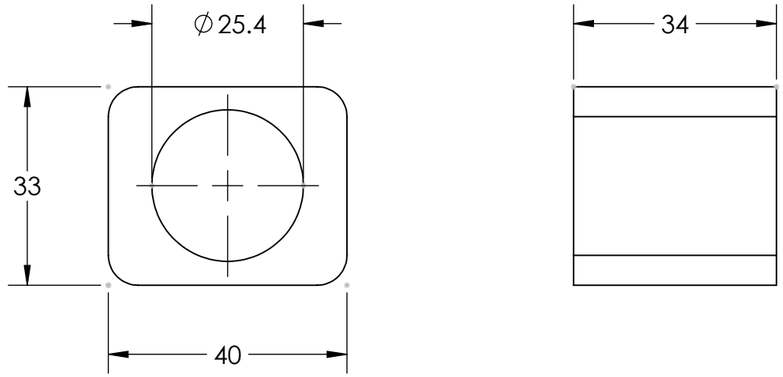
SCALE: 1:10 WEIGHT: SHEET 3 OF 5

SOLIDWORKS Educational Product. For Instructional Use Only.

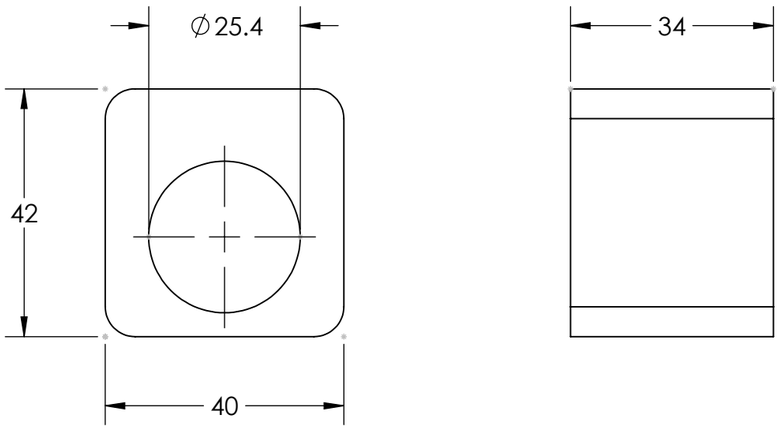
MOTOR MOUNT (PA6-CF), QTY 4 1:1



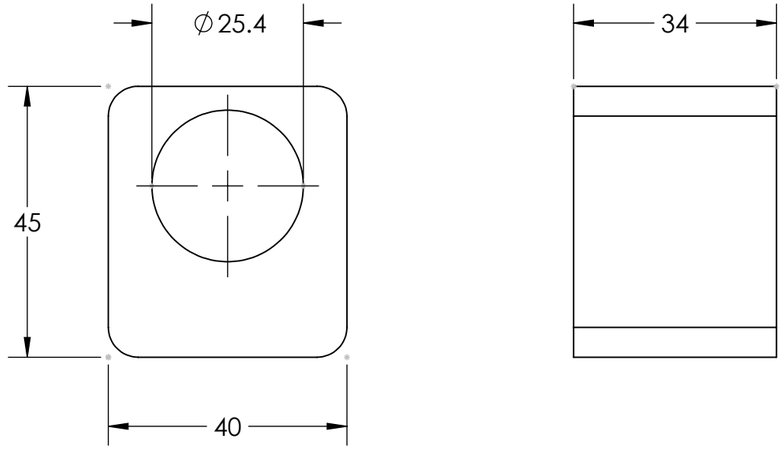
BOOM CLAMP, FRONT (PETG), QTY 2 1:1



BOOM CLAMP, REAR (PETG), QTY 2 1:1



GEAR CLAMP (PETG), QTY 4 1:1



NOTES
1. BORE 25.4 mm FOR THE 1.000 in BOOM: PRINT 0.2 mm UNDER, REAM TO FIT.
2. MOTOR MOUNT IN PA6-CF (HDT 173 °C): THE MOTOR CAN REACH ABOUT 73 °C IN A 90 s HOVER (ESTIMATE).
3. CLAMP TOPS MEET THE WING LOWER SURFACE. M3 HEAT-SET INSERTS FOR THE CLAMP SCREWS.

DESIGN-BUILD-FLY
STUDENT CAPSTONE PROJECT, UC BERKELEY.
CONCEPT GEOMETRY FOR DESIGN REVIEW, GENERATED FROM analysis/results.json AND cad/params.txt (v0.3.1).
NOT RELEASED FOR MANUFACTURE.

NEXT ASSY	USED ON
APPLICATION	

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETRES
GENERAL TOLERANCES (EST):
LINEAR X ±1 X.X ±0.5
ANGULAR ±1°
PRINTED PARTS ±0.3
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL: SEE PARTS LIST
FINISH: SEE PARTS LIST
DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	DBF	2026-10-18
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

DESIGN-BUILD-FLY

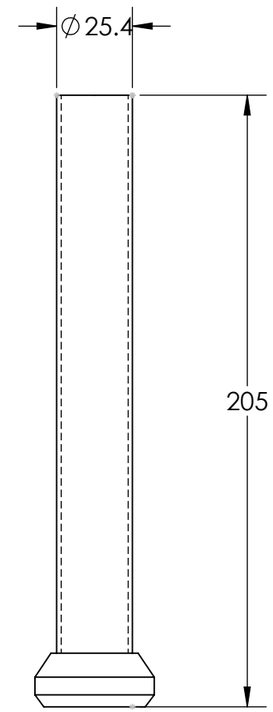
TITLE:
PRINTED PARTS
MOUNTS AND CLAMPS

SIZE	DWG. NO.	REV
B	DBF-DRW-001	0.3.1

SCALE: 1:1 | WEIGHT: | SHEET 4 OF 5

SOLIDWORKS Educational Product. For Instructional Use Only.

ITEM	PART	STOCK (6061-T6)	CUT LENGTH mm	QTY
1	Wing spar socket	1.000 x 0.058 in	800	2
2	Wing joiner	0.875 x 0.058 in	760	1
3	Front boom	1.000 x 0.035 in	955	2
4	Rear boom stub	1.000 x 0.035 in	516	2
5	Landing leg	1.000 x 0.058 in	205	4



LANDING LEG 1:2

NOTES
1. LEGS ARE CUT FROM THE SPAR-SOCKET TUBE OFFCUT:
   2 x 800 + 4 x 205 = 2420 mm OF A 96 in (2438 mm) LENGTH.
2. DEBURR ALL CUT ENDS. JOINER 0.23 mm DIAMETRAL
   CLEARANCE IN THE SOCKETS.
3. LENGTHS FROM cad/params.txt (v0.3.1).

DESIGN-BUILD-FLY

STUDENT CAPSTONE PROJECT, UC BERKELEY.
CONCEPT GEOMETRY FOR DESIGN REVIEW,
GENERATED FROM analysis/results.json
AND cad/params.txt (v0.3.1).
NOT RELEASED FOR MANUFACTURE.

NEXT ASSY	USED ON	
APPLICATION		

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN MILLIMETRES
GENERAL TOLERANCES (EST):
LINEAR X ±1 X.X ±0.5
ANGULAR ±1°
PRINTED PARTS ±0.3

INTERPRET GEOMETRIC TOLERANCING PER:

MATERIAL
SEE PARTS LIST

FINISH
SEE PARTS LIST

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	DBF	2026-10-18
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		

COMMENTS:

DESIGN-BUILD-FLY

TITLE:
TUBE CUT LIST
AND LANDING LEG

SIZE	DWG. NO.	REV
B	DBF-DRW-001	0.3.1

SCALE: 1:2	WEIGHT:	SHEET 5 OF 5

SOLIDWORKS Educational Product. For Instructional Use Only.